



The epidermal touch panel is capable of detecting motions such as tapping, holding, dragging and swiping (Image courtesy: www.toonts.com)

electrostatic field. Pressing on it closes the circuit, allowing the current to flow to the touch point. Current meters at each corner pick up the signals and transmit these to a separate controller board that connects to a computer.

World's first robot therapist helps relieve muscle strains

A Singapore based company, AiTreat, has developed a robot that can perform physiotherapy massages for muscle strains and injuries. Expert Manipulative Massage Automation, or EMMA, a robotic arm with a 3D-printed massage tip, can deliver massage therapy for sports injuries and muscular pains.



EMMA has already been used to treat Singapore's national basketball team using acupoint therapy (Image courtesy: www.economictimes.com)

EMMA is designed as a clinically precise tool to carry out treatment automatically as per a physiotherapist's prescription. It brings scientific data with a user-friendly interference and guidelines for pain management. It also helps medical practitioners carry out research and scientifically demonstrate to the patients the progress in treatment.

The robot has a single robotic arm that performs highly-articulate movements, a 3D stereoscopic camera for vision and a customised fully-rotatable 3D-printed massage tip. EMMA is also equipped with built-in pressure sensors to ensure safety and comfort of patients when performing pedal exercises.

Transparent circuitry to power smart windows

Scientists at King Abdullah University of Science & Technology (KAUST), Saudi Arabia, have found a way to make transparent transistors and other essential components of electronic circuitry, an advance that may lead to displays on car windscreens, transparent television sets and smart windows.



A transparent transistor for displays based on indium zinc oxide can dramatically increase performance (Image courtesy: kaust.edu.sa)

Indium tin oxide (ITO) is the current material of choice for electronics because it combines optical transparency with electrical conductivity. Its use ranges from touch-sensitive smartphone screens to light-harvesting solar panels. Indium is in short supply, and as demand increases for ITO-containing devices, so does the price of indium.

A low-cost ITO alternative is a transparent material known as aluminium-doped zinc oxide (AZO).

Researchers have used a high-precision technology called atomic layer deposition, a process in which the circuitry is built with a single layer of atoms at a time. Volatile vapours of aluminium and zinc in the form of trimethyl-aluminium and diethyl-zinc are alternately introduced onto the transparent substrate, where these adhere to the surface in a single layer before reacting in situ to form AZO.

Mind-reading car on road

Softbank founder Masayoshi Son has announced at an event in Tokyo that he is working with auto-giant Honda to come up with a car that can both talk and gauge human emotions.

Just like the talking car in the vintage series *Knight Rider*, the technology being developed will allow a vehicle to process natural language through cameras and in-car sensors, along with assessing the driver's emotions and engaging in friendly banter, in addition to offering assistance.

If such a level of interactivity is achieved, it could help in parking or keeping the driver engaged during long trips, as well as take over control when the driver is sleepy or not in senses.

Softbank's Pepper robot, recently powered by IBM's Watson AI, may have a role to play in this as well, Son added.